

Pea (Matar) Farming Guide for Pakistan

A complete crop guide for farmers, students, and agriculture advisors

1. Introduction

Pea, locally known as Matar, is a popular winter vegetable and pulse crop grown extensively across Pakistan for both fresh vegetable use and dry grain production. It is valued for its sweet-tasting green pods, which are widely used in home cooking, and its dry seeds, which are used in daal and other dishes. Pea is a cool-season legume crop, making it an important part of Pakistan's winter cropping system, especially in areas surrounding major cities where vegetable demand is high. Like other legumes, pea also improves soil fertility through nitrogen fixation, making it beneficial in rotation with cereal crops. Both garden pea (grown for fresh green peas) and field pea (grown mainly for dry seed) are cultivated in different parts of the country depending on local market demand.

2. Scientific Name

The scientific name of pea is *Pisum sativum*, belonging to the family Fabaceae (legume family). In Pakistan it is commonly known as Matar in Urdu and most regional languages.

3. Importance in Pakistan

Pea holds significant importance in Pakistan's winter vegetable economy, providing fresh green pods for household consumption, restaurants, and processing industries such as frozen food production. It is also an important source of income for vegetable growers around urban centers, where demand for fresh peas is consistently high during the winter season. As a legume, pea contributes to soil fertility improvement through nitrogen fixation, benefiting subsequent crops in rotation, typically wheat or maize. Dry pea seed is also used in various food preparations and as animal feed in some areas. The crop's relatively short duration and good market value make it an attractive option for smallholder vegetable farmers.

4. Major Producing Provinces and Districts

Pea cultivation is spread across Pakistan, with concentration near major urban and cooler areas. - **Punjab:** Major pea-growing districts include Kasur, Sheikhupura, Okara, Sahiwal, and areas around Lahore and Faisalabad, mostly for fresh vegetable markets. - **Khyber Pakhtunkhwa:** Swat, Mansehra, Abbottabad, and Peshawar valley are important pea-growing areas, benefiting from cooler climates suited to pea cultivation. - **Sindh:** Grown on a smaller scale in cooler winter months around Karachi and Hyderabad for fresh vegetable markets. - **Balochistan:** Quetta and surrounding highland areas grow pea, particularly benefiting from the cooler climate for quality pod production. Punjab remains the largest producer due to its extensive vegetable farming around major cities.

5. Climate Requirements

Pea is a cool-season crop that grows best in temperatures ranging from 10°C to 20°C, with optimal growth occurring around 15°C to 18°C. It is sensitive to hot weather, especially during flowering and pod formation, as high temperatures above 25°C can cause poor pod set and reduced quality. Pea can tolerate light frost once established, though young seedlings are more vulnerable to severe cold. It requires moderate moisture throughout its growth cycle but does not tolerate waterlogging, making well-timed irrigation important in areas without adequate winter rainfall.

6. Soil Requirements

Pea grows best in well-drained, fertile loamy soils rich in organic matter, though it can also be grown successfully in sandy loam soils with proper fertility management. Heavy clay soils with poor drainage are not suitable, as pea roots are sensitive to waterlogging and root rot. The ideal soil pH range for pea is between 6.0 and 7.5, with slightly acidic to neutral soils generally preferred for best nodulation and growth. Good soil fertility, supported by organic matter, contributes significantly to healthy pod development and yield.

7. Land Preparation

Proper land preparation is essential for good pea establishment and yield. The field should be ploughed two to three times to achieve a fine, well-pulverized seedbed, followed by planking to level the field and ensure uniform irrigation. Incorporating well-rotted farmyard manure during land preparation significantly improves soil fertility and structure, benefiting pea growth. In areas prone to waterlogging, raised beds or proper drainage channels should be prepared to prevent water accumulation, which can severely damage pea roots.

8. Recommended Varieties (Pakistan)

Several pea varieties are recommended for cultivation in different parts of Pakistan. - **Meteor:** A widely grown early-maturing variety popular for fresh vegetable markets. - **Climax:** Known for good pod size and yield, commonly grown in Punjab. - **Green Feast:** Popular for its sweet taste and good market appeal. - **Bonneville:** Suited to cooler highland areas such as Swat and Quetta. - **Local field pea varieties:** Grown in some areas mainly for dry seed production. Farmers should select varieties based on local climate, market preference (fresh pods versus dry seed), and availability of certified seed from local nurseries or seed suppliers.

9. Seed Rate

The recommended seed rate for pea depends on variety type and sowing method. - For garden pea (fresh vegetable type) sown in rows: 25 to 30 kilograms per acre. - For field pea (dry seed type): 20 to 25 kilograms per acre. - For dense sowing in small kitchen garden plots: slightly higher rates may be used to ensure a full stand. Using good quality, disease-free seed with high germination percentage helps achieve uniform crop stands and better yield.

10. Seed Treatment

Before sowing, pea seed should be treated with a fungicide such as Thiram or Carbendazim to protect against seed-borne diseases and damping-off in young seedlings. As a legume, pea seed benefits from treatment with Rhizobium bacterial inoculant, which improves nitrogen fixation, particularly in fields where pea or other legumes have not been grown recently. Seed should be handled carefully during treatment to avoid damaging the delicate seed coat, and treated seed should be sown promptly for best results.

11. Sowing Time (Province-wise)

- **Punjab:** Pea is typically sown from October to November for the main winter crop, with early varieties sometimes sown in September in cooler areas.
- **Khyber Pakhtunkhwa:** In highland areas like Swat and Abbottabad, sowing can extend into December due to the cooler climate, while lower areas follow a schedule similar to Punjab.
- **Sindh:** Sowing generally takes place from October to November, taking advantage of the relatively cooler winter months.
- **Balochistan:** In Quetta and surrounding highlands, sowing is often done slightly later, from October to November, due to the region's cool climate. Correct sowing timing is important to ensure flowering and pod development occur during the cooler months, avoiding heat stress that can reduce pod quality.

12. Sowing Methods

Pea can be sown using different methods depending on the scale of cultivation. - **Line sowing (drilling):** The most common method for commercial vegetable production, with row spacing of about 30 centimeters and plant spacing of 5 to 8 centimeters within rows. - **Broadcasting:** Sometimes used in smaller plots, though it results in uneven spacing and can make later management more difficult. - **Sowing on ridges:** Used in areas with heavier soils or higher risk of waterlogging, to improve drainage around the root zone. Proper spacing is important to allow good air circulation, which helps reduce disease pressure and supports healthy pod development.

13. Fertilizer Recommendations (NPK)

Pea requires moderate fertilizer inputs, given its nitrogen-fixing ability, but responds well to phosphorus and potassium. - **Nitrogen (N):** 15 to 20 kilograms per acre, applied as a starter dose at sowing to support early growth before nodulation becomes fully effective. - **Phosphorus (P2O5):** 40 to 50 kilograms per acre, applied fully at sowing time to support root and nodule development. - **Potassium (K2O):** 25 to 30 kilograms per acre, important for pod quality and overall plant vigor. Incorporating well-rotted farmyard manure along with chemical fertilizer further improves soil structure and nutrient availability for pea cultivation.

14. Irrigation Schedule

Pea generally requires four to five irrigations during its growing season under irrigated conditions, with critical stages being flowering and pod filling, when water shortage can significantly reduce yield and pod quality. The first irrigation is typically given soon after sowing to ensure good germination, followed by irrigations at regular intervals of about 12 to 15 days, adjusted based on rainfall and soil moisture. Waterlogging must be avoided at all stages, as pea roots are highly sensitive to excess moisture, which can lead to root rot and reduced yield.

15. Weed Management

Weeds can compete significantly with pea plants, especially during the first four to five weeks after sowing when the crop is still establishing. Effective weed control includes one or two hand weeding or hoeings during this early critical period, combined with careful shallow cultivation to avoid damaging the shallow root system of pea plants. Pre-emergence herbicides can be used in larger commercial fields, applied according to label instructions, though manual weeding remains common in smaller vegetable farms. Mulching is also sometimes used in vegetable-focused pea cultivation to suppress weeds and conserve soil moisture.

16. Insect Pests

Several insect pests can affect pea crops in Pakistan. - **Aphids:** Suck sap from leaves and stems, weakening plants and potentially spreading viral diseases. - **Pea pod borer:** Larvae feed inside developing pods, directly damaging seeds and reducing marketable pod quality. - **Leaf miner:** Creates visible tunnels in leaves, reducing photosynthetic capacity. - **Whitefly:** Can build up in warmer conditions, weakening plants and transmitting viral diseases. Regular field monitoring and timely application of recommended insecticides, used only when necessary, help protect crop quality while minimizing chemical residues on this widely consumed vegetable crop.

17. Diseases

Pea crops in Pakistan can be affected by various diseases, particularly under humid conditions. - **Powdery mildew:** Causes a white, powdery coating on leaves and pods, common in warmer, drier conditions later in the season. - **Rust:** Causes reddish-brown pustules on leaves, reducing photosynthesis and overall vigor. - **Root rot:** Affects plants in poorly drained or waterlogged soils, causing wilting and plant death. - **Downy mildew:** Causes yellowish patches on the upper leaf surface with grayish fungal growth underneath, particularly in cool, humid weather. Using resistant varieties where available, ensuring proper field drainage, and practicing crop rotation help manage these diseases effectively.

18. Deficiency Symptoms

Nutrient deficiencies in pea plants present identifiable symptoms. - **Nitrogen deficiency:** General yellowing of older leaves, though typically less severe once nodulation is well established. - **Phosphorus deficiency:** Dark green to purplish leaf coloration, along with poor root and nodule development and delayed flowering. - **Potassium deficiency:** Yellowing and scorching along leaf margins, with weaker stems and reduced pod quality. - **Calcium deficiency:** Can cause poor pod development and blossom-end issues in severe cases. - **Boron deficiency:** May cause hollow or malformed pods and stunted growth points. Timely correction of these deficiencies through balanced fertilization supports healthy plant growth and good quality pod production.

19. Harvesting

For fresh vegetable use, pea pods are harvested when they are well-filled but still tender and green, generally 60 to 90 days after sowing depending on variety, with harvesting done by hand-picking pods every few days as they reach the right maturity stage. For dry seed production, pea plants are allowed to mature fully until pods turn brown and dry, at which point plants are cut and left to dry further before threshing. Timely and regular picking is important for fresh market pea to maintain quality and encourage continued pod production from the plant over the harvesting period.

20. Storage

Fresh green pea pods have a short shelf life and should ideally be sold or consumed soon after harvest, though they can be stored briefly under cool, humid conditions or refrigerated to maintain quality for a few days. For dry pea seed, proper drying to about 10 to 12 percent moisture content is essential before storage, to prevent insect infestation and mold growth. Dry seed should be stored in clean, dry, well-ventilated containers, protected from pests such as pulse beetles, and kept in a cool, dry location for extended storage life.

21. Expected Yield in Pakistan

For fresh green pea (in-pod), average yield in Pakistan ranges from about 40 to 60 maunds per acre (1,500 to 2,250 kilograms per acre), depending on variety, management, and local growing conditions. For dry pea seed production, yields typically range from about 10 to 15 maunds per acre (375 to 560 kilograms per acre). Actual yield depends significantly on variety selection, sowing time, irrigation availability, and effective pest and disease management throughout the season.

22. Government Recommendations

Provincial horticulture and agriculture departments, along with research institutes such as the Vegetable Research Institute at Ayub Agricultural Research Institute (AARI) Faisalabad, provide regular recommendations for pea cultivation. These typically include guidance on using certified, disease-free seed of recommended varieties, following proper sowing schedules suited to local climate, and applying balanced fertilizer based on soil testing where available. Farmers are also encouraged to practice crop rotation with cereals to maintain soil health and to seek updated advice from local agriculture extension offices regarding pest management and market-oriented variety selection.

23. Modern Farming Practices

Vegetable farmers growing pea in Pakistan are increasingly adopting improved practices such as line sowing with proper spacing for better yield and easier crop management, use of mulching to conserve moisture and suppress weeds, and drip irrigation systems in some commercial vegetable farms to improve water use efficiency. Improved post-harvest handling, including prompt cooling and careful packing of fresh pea pods, is also becoming more common among farmers supplying urban markets and processing industries.

24. Precision Agriculture

Precision agriculture practices are gradually being introduced into pea cultivation, particularly among commercial vegetable growers near major cities. These include the use of soil testing to guide precise fertilizer application, drip irrigation systems combined with moisture sensors to optimize water use during critical flowering and pod-filling stages, and greenhouse or tunnel cultivation in some areas to extend the growing season and protect crops from unfavorable weather. While such technologies remain more common among larger commercial operations, growing awareness through agricultural extension services is helping smaller farmers gradually adopt similar practices.

25. Sustainable Farming

Pea supports sustainable farming through its nitrogen-fixing ability, which reduces dependence on synthetic nitrogen fertilizer and improves soil fertility for subsequent crops in rotation, typically wheat or maize. Sustainable practices encouraged in pea cultivation include crop rotation to break pest and disease cycles, use of organic matter such as farmyard manure to improve soil health, and integrated pest management to reduce reliance on chemical pesticides, which is particularly important for a vegetable crop consumed fresh by households. Efficient irrigation practices, including proper scheduling to avoid both water stress and waterlogging, further support sustainable water use in pea cultivation.

26. Summary Note

This guide reflects pea (Matar) farming practices specific to Pakistan, covering major growing regions in Punjab, Khyber Pakhtunkhwa, Sindh, and Balochistan. Farmers should confirm current variety recommendations, seed sources, and sowing schedules with their local agriculture extension office, as these are updated periodically based on new research and market needs.